

TEST PLAN DOCUMENT

Project : online internship and placement preparation portal

Naming convention : UT = Unit Testing , IT = Integration Testing , ST = System Testing.

Functional Requirements Test Cases

FR-001 — Student Registration & Login

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_01	User Authentication	Verify login with valid registered credentials	Student account already registered	1: Navigate to Login page 2: Enter valid username 3: Enter valid password 4: Click Login	Username: student1@test.com Password: Pass@123	Login successful; student redirected to profile dashboard		
UT_02	User Authentication	Verify login is rejected with an incorrect password	Student account already registered	1: Navigate to Login page 2: Enter valid username 3: Enter incorrect password 4: Click Login	Username: student1@test.com Password: Wrong@123	Login rejected; error message "Invalid username or password" displayed		
UT_03	User Authentication	Verify login is rejected for a non-existent username	None	1: Navigate to Login page 2: Enter unregistered username 3: Enter any password 4: Click Login	Username: notexist@test.com Password: Pass@123	Login rejected; error message "Invalid username or password" displayed		
UT_04	User Registration	Verify registration form validates mandatory empty fields	User is on Registration page	1: Leave Name, Email and Password fields blank 2: Click Register	Name: (blank) Email: (blank) Password: (blank)	Form displays field-level validation errors; registration not submitted		
IT_01	User Registration + Login	Verify a newly registered student can immediately log in with the same credentials	Email not previously used	1: Complete registration form with new details 2: Click Register 3: Navigate to Login page 4: Enter same credentials 5: Click Login	Email: newstudent@test.com Password: New@123	Registration successful; subsequent login with same credentials succeeds		
IT_02	User Registration	Verify registration is blocked when the email already exists	Account with given email already registered	1: Navigate to Registration page 2: Enter an already-registered email	Email: student1@test.com (already registered)	Error message "Email already exists" displayed; duplicate account not created		

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
				3: Fill remaining fields 4: Click Register				
IT_03	User Authentication	Verify user session persists while navigating across pages after login	Student logged in successfully	1: Log in 2: Navigate to Practice module 3: Navigate to Internships module 4: Navigate to Analytics module	N/A	User remains authenticated on every page; no re-login prompted		
ST_01	User Authentication	End-to-end: new student registers, logs in and reaches profile dashboard	None	1: Register new account 2: Log in with new credentials 3: Verify landing on profile dashboard	Email: e2euser@test.com Password: E2e@123	Full flow completes successfully; dashboard loads with student profile details		
ST_02	User Authentication	Verify expired/logged-out session requires re-authentication to access protected pages	Student previously logged in	1: Log out (or allow session to expire) 2: Attempt to directly access dashboard URL	N/A	User redirected to Login page; protected content not accessible		

FR-002 — Internship Listings & Application

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_05	Internship Listings	Verify admin-posted internship listings are displayed to students	At least one active listing posted by admin	1: Log in as student 2: Navigate to Internships tab	N/A	All active listings displayed with title, company and deadline		
UT_06	Internship Application	Verify student can apply to an internship listing	Student logged in; at least one open listing exists	1: Open an internship listing 2: Click Apply 3: Confirm application	Listing: "Backend Intern - Acme Corp"	Application recorded; confirmation message displayed		
UT_07	Internship Application	Verify duplicate application to the same listing is blocked	Student has already applied to the listing	1: Open the same listing already applied to 2: Click Apply again	Listing: "Backend Intern - Acme Corp"	Error message "You have already applied to this internship" displayed		
IT_04	Internship Application	Verify an applied internship appears under "My Applications"	Student has just applied to a listing	1: Apply to a listing 2: Navigate to "My Applications"	Listing: "Data Analyst Intern - Beta Inc"	Listing appears under My Applications with correct status (e.g., Applied)		
IT_05	Internship Listings	Verify a newly posted admin listing becomes visible to students	Admin has access to post listings	1: Admin posts a new internship listing	New listing: "QA Intern - Gamma Ltd"	New listing visible in student's internship list without delay		

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
				2: Student refreshes/reopens Internships tab				
IT_06	Internship Application	Verify a student can apply to multiple different listings	At least two open listings exist	1: Apply to Listing A 2: Apply to Listing B 3: Navigate to "My Applications"	Listing A, Listing B	Both applications recorded and visible independently under My Applications		
ST_03	Internship Module	End-to-end: admin posts listing, student views, applies, and tracks status	Admin and student accounts available	1: Admin posts new listing 2: Student views listing 3: Student applies 4: Student checks status under My Applications	Listing: "Frontend Intern - Delta Co"	Full flow completes; application status correctly tracked end to end		
ST_04	Internship Listings	Verify students cannot apply to a listing past its deadline	A listing exists with an expired deadline	1: Open a listing whose deadline has passed 2: Attempt to click Apply	Listing deadline: past date	Apply option disabled/hidden, or error "Application window closed" shown		

FR-003 — Practice Question Bank (Aptitude / Technical MCQ / Coding)

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_08	Practice - Aptitude	Verify selecting Aptitude category returns relevant questions	Question bank contains Aptitude questions	1: Navigate to Practice module 2: Select "Aptitude" category	Category: Aptitude	Aptitude questions displayed for the student to attempt		
UT_09	Practice - Technical MCQ	Verify selecting Technical MCQ category returns relevant questions	Question bank contains Technical MCQs	1: Navigate to Practice module 2: Select "Technical MCQ" category	Category: Technical MCQ	Technical MCQ questions displayed for the student to attempt		
UT_10	Practice - Coding	Verify selecting Coding category returns relevant coding problems	Question bank contains coding problems	1: Navigate to Practice module 2: Select "Coding" category	Category: Coding	Coding problems displayed for the student to attempt		
UT_11	Practice Scoring	Verify submitted practice answers are scored correctly	Student has attempted a set of practice questions	1: Answer all displayed questions 2: Click Submit	5 MCQ answers (mix of correct/incorrect)	Scored result displayed matching number of correct answers		
IT_07	Practice - Question Bank	Verify appropriate message shown when a category has no available questions	A category exists with zero questions configured	1: Navigate to Practice module 2: Select the empty category	Category: "Advanced DSA" (no questions configured)	"No questions found" message displayed; no application error/crash		

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
IT_08	Practice - History	Verify a completed practice attempt is logged to student history	Student completes a practice set	1: Complete a practice set in any category 2: Navigate to practice history/analytics	Category: Aptitude	Attempt appears in student's practice history with date and score		
ST_05	Practice Module	End-to-end: select category, answer all questions, and view scored result	None	1: Select a category 2: Attempt all questions 3: Submit 4: View result screen	Category: Technical MCQ	Complete flow works; scored result and category shown correctly		
ST_06	Practice Module	Verify switching category mid-session loads a fresh question set without mixing	Student has started a practice session in one category	1: Start practice in "Aptitude" 2: Switch to "Technical MCQ" before submitting 3: Observe question set	Category switch: Aptitude -> Technical MCQ	Only Technical MCQ questions shown; no Aptitude questions carried over		

FR-004 — Timed Mock Tests & Result Analysis

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_12	Mock Test	Verify a timed mock test starts with a visible countdown timer	Student logged in	1: Navigate to Mock Test module 2: Click Start Mock Test	Test: "Full Aptitude Mock - 30 min"	Timer starts counting down; questions displayed		
UT_13	Mock Test	Verify manual submission before the time limit is accepted	Mock test in progress, time remaining	1: Answer required questions 2: Click Submit before timer expires	N/A	Test submitted successfully; score screen displayed		
UT_14	Mock Test	Verify the test auto-submits when the time limit expires	Mock test in progress	1: Start mock test 2: Do not submit; allow timer to reach 00:00	Test duration: e.g., 1 min (for testing)	Test auto-submits at time limit; score screen shown without manual action		
UT_15	Mock Test	Verify detailed correct/incorrect answer breakdown is shown after submission	Mock test submitted (manually or auto)	1: Submit mock test 2: Open detailed analysis screen	N/A	Each question marked correct/incorrect with the correct answer shown		
IT_09	Mock Test - History	Verify a completed mock test result is stored and retrievable later	Student completes a mock test	1: Complete and submit a mock test 2: Navigate away from results page 3: Return via Test History	N/A	Same result (score and breakdown) retrievable from history		

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
IT_10	Mock Test	Verify the test timer is not reset by a page refresh mid-test	Mock test in progress	1: Start mock test 2: Note remaining time 3: Refresh the page 4: Compare remaining time	N/A	Timer continues from correct elapsed time; not reset to full duration		
ST_07	Mock Test Module	End-to-end: take a full mock test, let it auto-submit, and view detailed analysis	None	1: Start mock test 2: Answer some questions 3: Allow timer to expire 4: View detailed analysis	Test: "Full Aptitude Mock - 30 min"	Auto-submit occurs; score and correct/incorrect breakdown displayed accurately		
ST_08	Mock Test Module	Verify results of multiple mock tests taken back-to-back are recorded distinctly	Student completes at least two mock tests in one session	1: Complete Mock Test A and submit 2: Immediately start and complete Mock Test B 3: Check Test History	Mock Test A, Mock Test B	Both results recorded separately with correct scores; no data overlap		

FR-005 — Personal Analytics Summary

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
UT_16	Analytics Dashboard	Verify performance trend chart is displayed on the analytics dashboard	Student has at least one test/practice attempt	1: Navigate to Analytics dashboard	N/A	Trend chart of past scores displayed		
UT_17	Analytics Dashboard	Verify strengths/weaknesses summary is shown per category	Student has attempts across multiple categories	1: Navigate to Analytics dashboard 2: View strengths/weaknesses section	N/A	Categorized summary (e.g., strong in Aptitude, weak in Coding) displayed		
UT_18	Analytics Dashboard	Verify analytics data is isolated to the logged-in student only	Two student accounts exist with different histories	1: Log in as Student A 2: View Analytics dashboard 3: Log out and log in as Student B 4: View Analytics dashboard	Student A, Student B accounts	Each student sees only their own performance data; no cross-visibility		
IT_11	Analytics Dashboard	Verify the dashboard updates immediately after a new test/practice attempt	Student has an existing analytics history	1: Note current dashboard values 2: Complete a new practice/mock test attempt 3: Return to Analytics dashboard	New attempt score	Dashboard reflects updated scores/trend; no stale cached data shown		

Test Case ID	Name of Module	Test Case Description	Pre-conditions	Test Steps	Test Data	Expected Results	Actual Result	Test Result
IT_12	Analytics Dashboard	Verify aggregated analytics are calculated accurately across multiple attempts	Student has 3 or more recorded attempts with known scores	1: Record scores of 3 attempts manually 2: Open Analytics dashboard 3: Compare displayed average/trend with manual calculation	Attempt scores: 60%, 75%, 90%	Displayed aggregate matches manually calculated average/trend		
ST_09	Analytics Dashboard	Verify a new student with no attempts sees an empty state, not a broken dashboard	Newly registered student with zero attempts	1: Log in as new student 2: Navigate to Analytics dashboard	N/A	"No data yet" message displayed; no errors or broken charts		
ST_10	Analytics Dashboard	End-to-end: complete several tests/practice sessions and verify dashboard reflects combined performance	None	1: Complete 2 practice sessions and 1 mock test 2: Navigate to Analytics dashboard 3: Verify trend and strengths/weaknesses reflect all attempts	2 practice attempts + 1 mock test	Dashboard accurately aggregates and displays combined performance data		
ST_11	Analytics Dashboard	Verify strengths/weaknesses summary updates correctly when a weak category improves over time	Student has multiple past attempts in a category showing weakness	1: Note current weak category on dashboard 2: Complete several new practice attempts scoring well in that category 3: Refresh Analytics dashboard	New attempts: 3 high-scoring attempts in previously weak category	Dashboard updates the category from 'weak' to 'improved/strong' based on latest trend		